INCH-POUND
MIL-E-24786(SH)
21 March 1994

MILITARY SPECIFICATION

ELECTROCHEMICAL HYPOCHLORITE GENERATION (EHG) SYSTEMS FOR DOCKSIDE SERVICE

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, dockside electrochemical hypochlorite generation (EHG) systems.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-C-596 - Connector, Plug, Receptacle and Cable Outlet, Electrical Power General Specification for.
TT-E-490 - Enamel, Silicone, Alkyd, Copolymer, Semigloss (for exterior and interior non-residential use).
WW-C-621 - Coupling Assembly, Hose.
ZZ-H-561 - Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 03R42, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4610
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MIL-P-15024 - Plates, Tags and Bands for Identification Equipment.
MIL-P-15024/10 - Nameplates, Ordalt Plates and Idenification Plates.
MIL-E-16400 - Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore; General Specification for.
MIL-E-17555 - Electronic and Electrical Equipment Accessories and Provisioned Items (Repair Ports); Packaging of.
MIL-P-24441 - Paint, Epoxy-Polyamide, General Specification for.

STANDARDS

MILITARY

MIL-STD-889 - Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

United States Government Printing Office
GPO - Style Manual.

(Copies of the GPO Style Manual are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issue of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.3).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
Enclosures for Electrical Equipment (1000 Volts Maximum).

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

INSTITUTE FOR ELECTRICAL AND ELECTRONIC ENGINEERS
IEEE-45 - Recommended Practice for Electrical Installation on Shipboard.

(Application for copies should be addressed to the Institute for Electrical and Electronic Engineers, 445 Hose Lane, P.O. Box 1331, Piscataway, N.J. 08855.)

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STEEL STRUCTURES PAINTING COUNCIL (SSPC)
SSPC-SP-5 - Steel Structures Painting Council Manual, Systems and Specifications, Volume II, Chapter 2.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburg, PA 15213-2683.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Description. The EHG system shall operate continuously or intermittently under timer control to supply hypochlorite solution via a hose and dispersal unit to up to 2 ship seawater intakes individually operating between 200 gallons per minute (gpm) and 6,000 gpm. Only one dispersal unit shall be in operation when ship seawater intake flow rates are greater than 2000 gpm. The hypochlorite solution shall be injected into the ship seawater system by a dispersal system magnetically attached to the ships hull. The dispersal systems shall provide hypochlorite which shall dilute as it enters the ship seawater intake.

3.3 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the EHG systems to meet the performance requirements of this specification.

3.3.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.2 Painting. The steel components of the base plate, skid and trailer shall be abrasive blasted to near white metal in accordance with SSPC-SP-5 and coated with a 3-coat epoxy coating system in accordance with MIL-P-24441 prior to assembly. This shall be topcoated with two coats of haze gray silicone alkyd in accordance with TT-E-490.

3.4 Environmental conditions. The EHG systems shall perform as specified herein in a salt air environment with ambient air temperature from 32 to +125 degrees Fahrenheit (°F). The EHG systems shall operate using natural saltwater with salinity ranging from 15 parts per thousand (ppt) to 35 ppt with seawater temperature ranging from 50 to 90°F. The EHG shall operate with seawater supply pressures from 20 to 100 pounds per square inch gauge (psig) and shall be hydrostatically tested at 150 psig. Drainage shall be provided at low points in the system for storage in freezing weather.

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3.5 Automatic shutdown. In order to prevent environmental impact the system shall be designed for constant operation with automatic shutdown capabilities for the following conditions:

(a) low flow entering the generator (see 3.10.1.1);
(b) negligible ship seawater intake flow (see 3.9);
(c) high voltage (see 3.7.2).

During automatic shutdown, a warning light and audible signal shall come on, and the power supply output shall automatically shut off. In case of automatic shutdown, the unit shall require manual restart when the condition causing the shutdown is corrected.

3.6 Hypochlorite generator. The hypochlorite generator shall be an electrolytic sodium hypochlorite type. The amount of hypochlorite produced shall be governed by the amount of current passed through the cells. The current shall have variance from 1 to 100 percent of maximum designed output. The generator shall generate 1.5 pounds of equivalent chlorine per hour in seawater with salinity of 15 ppt and seawater temperature of 50°F. The generator feed flow rate shall be 26 gallons per minute. Temperature and pressure gages, control and sampling valves shall be provided as necessary to monitor the generator's operations.

3.6.1 Generator construction. The generator cells shall consist of anodes and cathodes made from commercially pure titanium (min 98.8 percent). The anodes shall be plated with pure platinum (min 99.0 percent) coating. The inlet and discharge points on the generator shall be electrically connected so that they are of the same electrical potential and shall be connected to ground. Insulators and other materials shall be made of polyvinyl chloride or glass reinforced plastic. The cells shall be designed to operate at 100 psig with a 26 gallon per minute flow rate and shall be hydrostatically tested to 150 psig.

3.6.2 Generator maintenance. The generator shall be designed such that disassembly or chemical cleaning is not required during a minimum of 2,200 hours of continuous service. Flow turbulence, flow velocity, automatic polarity reversals, or other methods which meet the 2,200 hour maintenance requirement are acceptable. At constant salinity and temperature, the current demand shall not increase more than 10 percent to generate 1.5 pounds of equivalent chlorine per hour at a constant 26 gpm. At full output, the generator shall continue to generate a minimum of 90 percent of the rated full output.

3.7 Power supply/controller. The power supply/controller shall be a high reliability saturable reactor type. It shall continuously maintain a preset current to the generator regulating hypochlorite output. The power supply/controller shall be able to maintain between 2% and 100% of total current output.

3.7.1 Construction. The power supply/controller shall be housed in a NEMA-3X enclosure as specified in NEMA Enclosures for Electrical Equipment.

3.7.2 Design. The power supply shall operate using 480 V +/- 10 percent, 3 phase, 60 Hz, input power during normal operation. Proper ventilation shall be provided to prevent overheating of the power supply. The design shall include a control panel with status lights, digital meters, switches, and a timer. The

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status lights shall show normal operation and alarm conditions. Red digital meters shall be provided on the indicator panel for monitoring total current and system voltage. Status lights shall be provided to show which ship seawater intake is flowing. Low voltage, high current DC shall be provided for the hypochlorite generator cells. An electric load center with 120 V plus or minus 10 percent, single phase, 60 Hz service outlets shall be available for a hypochlorite sensing system, a 1 horsepower seawater pump, and test equipment. Power shall be supplied to the pump outlet at the load center if the pump is required. The pump shall be operated by a switch at the load center if a pump is required. A phase reversal switch shall be wired into the pump control circuit. The power supply shall have input and output circuit breakers and a relay to interrupt power to the input of the generator. The power supply cord shall be in accordance with W-C-596. The sensors and load center shall have an insulation resistance of not less than 10 megohms.

3.7.3 Electrical Wiring. The electrical wiring necessary to connect all system components shall be provided. All wiring will meet IEEE-45 standards for marine environments. This requirement includes but is not limited to provisions regarding circuit breakers, wire gauge sizes, terminal block design/configurations and numbering.

3.8 Timing device. A solid state timing device with a minimum of eight-event (on-off) functions that allows hypochlorite generator operation at preset intervals during a 24 hour period shall be provided. The timer shall include manual and automatic modes; set, review, clear, clock buttons all with Red Digital controls. The timer shall have a minimum 30-day battery back-up for memory in the event that power to the timer is interrupted.

3.9 Ship seawater intake flow sensors. The ship seawater intake flow sensors shall be the miniature turbine or paddle type which generate an electrical signal when the turbine blades or paddle wheel is rotated by the water stream. The signal shall be transmitted to flow meters located on the main control panel of the power supply. The flow sensors shall be made of a corrosion resistant material (e.g. titanium, niobium). The flow sensors shall be located on each of the dispersal units (see 3.12) such that it sits flush on the ship seawater intake grating. The flow sensor shall be contained in a pipe so that natural cross currents and undertows will not be detected as ship seawater intake flow.

3.9.1 Dispersal unit control. The individual flow control units shall control release of hypochlorite solution to a dispersal unit through electrically operated motorized ball valves. These valves shall be open only if the ship seawater intake is operating at a minimum of 0.5 ft/sec. When two dispersal units are in use, the flow sensors shall control the two electrically operated motorized ball valves, located on the hypochlorite distribution piping, by directing the hypochlorite to the dispersal manifold attached to the active ship seawater intake operating at greater than 0.5 ft/sec. If there is negligible seawater flow (less than 0.5 ft/sec) through both of the ship seawater intakes, the generator shall automatically shutdown.

3.10 Piping systems. Piping for the EHG system shall be as follows.

3.10.1 Seawater supply piping. The hypochlorite generator, shall operate from a constantly pressurized source of seawater. This shall be accomplished by hooking up to a seawater fire main or a pump which pumps seawater from the harbor.

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A fire hose connector in accordance with WW-C-621 for seawater use shall be provided for hooking up seawater supply, as well as a seawater supply valve. All seawater inlet piping shall be 90-10 copper nickel alloy. All values shall be a compatable copper alloy (e.g. Bronze). A commercial seawater strainer shall be provided for the seawater inlet piping. The body shall be bronze with a MONEL or passivated CRES basket. The mesh size shall be half (or the next smaller size) of the chlorinator generation cell gap.

3.10.1.1 Low flow sensor. A low flow sensor shall be located in the inlet piping system prior to the generator. The flow sensor shall be made of a corrosion resistant material (e.g. titanium, Niobium). The flow sensor shall transmit a signal to a relay and an adjustable set-point control. The purpose of the relay is to activate a warning light, an audible signal, and shut down the generator in the event of abnormal low flow conditions of less than 20 gallons per minute.

3.10.2 Hypochlorite distribution piping. All hypochlorite distribution piping shall be made of schedule 80 PVC and Schedule 5 titanium (where strength is required). The piping shall be able to withstand regular jarring movements experienced when the chlorinator is moved. Two outlets shall be provided to allow two ship seawater intakes to receive hypochlorite. Each outlet shall have a shut-off valve, flow monitor, a motorized ball valve, and a hose connection. The hose connection shall be on the exterior of the enclosure for the dispersal manifold assembly and shall not protrude from the skid.

3.11 Hydrogen separator. Provisions shall be provided for separation of the hydrogen gas byproducts from the hypochlorite solution immediately as the hypochlorite exits from the generator. The hydrogen gas shall be automatically vented, without liquid loss, at a point above the unit enclosure.

3.12 Dispersal unit assembly. A dispersal unit assembly shall be provided for each outlet. The dispersal unit assembly shall consist of a 1-inch discharge hose (150 feet long) in accordance with ZZ-H-561, a titanium and PVC dispersal unit with attachment, and a flow sensor (see 3.9). The dispersal unit assembly shall mix the hypochlorite solution with the seawater entering the ship seawater intake. The dispersal unit shall be designed to provide the hypochlorite stream a six inch zone of mixing prior to the ship intake in order to protect the ship seawater intake from fouling. The flow sensor shall be located on the dispersal unit so as to not be in the flow of the hypochlorite solution. The dispersal unit shall have two 150 pound pull magnets used for attachment to the ship's hull. The dispersal unit shall have a pull cord which is cleated off at the hull used for disengaging the dispersal unit assembly. The use of divers is permitted for dispersal unit attachment, but not for disengagement. Accommodations to easily stow and remove the dispersal unit assembly within the equipment enclosure shall be provided. Hose racks shall be provided exterior to the enclosure on the trailer.

3.13 Equipment enclosure. A removable NEMA 4 enclosure shall be provided to house and protect the EHG system. The enclosure shall be used outdoors and shall be constructed of haze grey glass reinforced plastic with CRES 316 hardware. Access covers shall be removed by a single technician for servicing. Viewing windows shall be provided to allow monitoring EHG system operation without opening the enclosure. A vent shall be installed on the top of the enclosure in the event of a leak in the hydrogen separator. The enclosure shall have removable or hinged access panels which will allow easy access for monitoring, inspection, maintenance

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and adjustment of all EHG system components including internal access to electronic cabinets. The enclosure shall be mounted on the skid (see 3.14) to provide a self-contained, portable unit. Unless otherwise specified (see 6.2) the enclosure shall not be more than 6 ft. long by 4 ft. wide by 4 ft. tall.

3.14 Base plate and skid. The EHG system shall be fully portable and self-contained. The skid shall have fork lift tubes installed to allow movement by a forklift. A base plate and skid to suit the equipment layout and installation shall be provided. The skid and base plate shall be mounted onto a trailer capable of being towed behind a vehicle for the purpose of movement between local storage and the pier. The trailer shall be set up so that the unit will not tilt more than five degrees from vertical at any time. The trailer shall be equipped with a pintol hook. All chlorinator components shall be mounted on the skid within the unit enclosure. Any internally mounted components or equipment weighing 150 or more pounds shall be provided with eyebolts. The base plate, skid and trailer shall be coated in accordance with 3.3.2. Drain ports shall be provided in the skid to facilitate removal of any standing water.

3.15 Label plates.

3.15.1 Nameplates. Each unit shall be marked with a permanently affixed nameplate as specified in MIL-P-15024/10 made clearly visible to personnel. The nameplate shall have the following information as a minimum: Manufacturer's name and model designation, unit name, AC input current at rated output, AC input frequency, rated output.

3.15.2 Identification plates. All controls, indicating instruments, connections, and major components shall be appropriately labeled.

3.15.3 Instruction plates. Clear instructions shall be affixed to the equipment enclosure on how to operate the EHG system.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be conducted on 1 completed unit and shall consist of the examinations and tests specified in table I (see 6.5).

TABLE I. First article inspection requirements.

Description	First Article	Quality Conformance	Reference paragraph	
			Requirement	Inspection
General Examination	X	X	3.3	4.4.1
Major Components	X	X	3.3	4.4.2
Completed Units	X	X	3.3	4.4.3
Hypochlorite output	X	X	3.6	4.5.1
Electrode scaling tests	X		3.6	4.5.2
Leak test	X	X	3.4	4.5.3
Insulation resistance	X		3.7.2	4.5.4
Phase balance	X	X	3.7.2	4.5.5
Pump load center test	X		3.7.2	4.5.6
Timer controls	X	X	3.8	4.5.7
Flow meter calibration	X	X	3.10	4.5.8
Low flow alarm test	X		3.10.1.1	4.5.9
Flow sensors	X		3.9	4.5.10
Effectiveness of enclosure	X		3.10	4.5.11

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4.4 Quality conformance inspection.

4.4.1 General examination. The contractor shall ensure that the final assembly conforms to the prescribed requirements (see 6.3). Inspection procedures shall be as specified in MIL-E-16400. Each completed unit and items of major parts shall be subjected to the quality conformance inspections and tests specified in table I.

4.4.2 Major components. Examination shall be as follows:

(a) To determine conformance with the applicable specification and that the size, type or rating are as shown (see 6.3):

(1) Resistors and rheostats.
(2) Instruments, instrument transformers, shunts and other associated accessories.
(3) Motor starters and associated pushbuttons.
(4) Circuit breakers.
(5) Capacitors.
(6) Switches.
(7) Semiconductor devices.
(8) Terminal Boards.
(9) All laminated materials, such as glass mica glass (GMG), glass silicon glass (GSG); used for terminal boards, subpanels, etc.

(b) Electrical insulation tests shall be performed between all windings and each winding and the core.
(c) No black varnish, glyptol, paint, or other treatment was given to the insulation of reactors and transformers.
(d) The temperature rise of all reactors and transformers is as shown (see 6.3). This test shall be performed on the first article inspection.
(e) The equipment enclosures outline and mounting dimensions and material are as shown (see 6.3).

4.4.3 Completed unit. Examination shall be conducted to determine that:

(a) All components are mounted (see 6.3).
(b) All parts which require periodic adjustment or maintenance during the life of the equipment are accessible without removing other parts.
(c) All major components are marked with identifying symbols and those symbols agree (see 6.3).
(d) All terminals of all major components and terminal boards are marked with identifying letters and numbers, or both (see 6.3).
(e) Wire used for interconnecting components (see 6.3).
(f) All wiring is neatly formed into groups and neatly laced and supported or clamped so as to prevent chafing of the insulation due to vibration.
(g) Cable clamps used are of nonflammable material.
(h) There are no splices in any of the wires.
(i) Wire groups from hinged doors and panels are formed and clamped such that sharp bends do not occur with the panel or door in

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either the open or closed position.

(j) Sufficient slack in wiring is allowed so that the weight of the harness is not supported by the terminal connections and so that at least two half-inch replacements can be made if lugs are clipped off at the wire end.

(k) Cable entrance provisions (see 6.3).

(l) Sufficient space is provided to bring external cables through these cable entrances, support them within the enclosure, and make connections at the appropriate terminals.

(m) Wherever wires run through holes in metal partitions or the chassis, grommets are provided for mechanical protection.

(n) Wires are not bent around sharp corners which may injure the insulation.

(o) All wires to major components are connected by either bolts, terminal lugs, compression box terminals, or soldered connections.

(p) All bolted connections are provided with locking devices.

(q) Both ends of all wires are color-coded or marked with designations (see 6.3).

(r) Hinged doors and panels do not bind when opening.

(s) All identification and instruction plates are furnished (see 6.3).

(t) The location of the viewing windows allows full view of control panel with operating modes.

(u) The access panels allow easy access to equipment that will require inspection and maintenance.

4.5 Tests.

4.5.1 Hypochlorite generator production test. With the hypochlorite generator operating normally (26 gpm) in seawater at 15 ppt salinity and a temperature of 50°F and 90°F, the generator shall produce 1.5 pounds of equivalent chlorine per hour (± 10%). These tests shall be conducted after the generator has been running a minimum of 8 hours continuously.

4.5.2 Electrode scaling test. The hypochlorite generator shall be continuously operated at 1.5 pounds per hour chlorine output (operating with seawater that has relatively constant salinity and temperature) for a period of 2,200 hours. With a constant current setting the subject generator output shall not degrade more than 10 percent.

4.5.3 Leak test. With the hypochlorite generation system denergized, fill the system with water. Block the discharge ports. From an external source, increase the pressure to 150 psi. Hold for 15 minutes at 150 psi and then inspect the system for leaks. In addition, the hydrogen seperator shall have no liquid loss. All leaks are to be recorded and repaired by the contractor. All repaired units shall be retested in accordance with this paragraph to ensure no leakage exists.

4.5.4 Insulation resistance. The measurement of insulation resistance shall be made for the sensors and load centers. The resistance measuring equipment shall be similar and equivalent in accuracy to the General Radio Megohm Bridge, Type 544-B, 500 Volt D.C. with a range from 0.1 to 1,000,000 megohms. The insulation resistance shall not be less than 10 megohms when measured with all circuits cold.

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4.5.5 Phase balance. The unit with three-phase, three-wire ungrounded input shall be tested at 50 percent and full load for current balance in the A.C. lines. The A.C. line current of each phase shall be measured at the two load points. A variation between phases of more then +/- 20 percent at 50 percent load and +/- 10 percent at 100 percent load shall be cause for rejection.

4.5.6 Pump load center test. The pump load center on the power supply/ controller shall be tested to verify that power is supplied when the pump is required.

4.5.7 Timer control. For first article and quality conformance inspections, the timer shall be set for eight events (on-off). For first article inspection only, the chlorinator shall be run through two 24-hour cycles to verify correct operation of the timers. The chlorinator shall be turned off for a period of 29 days, after which time power will be returned and the chlorinator will be run through two 24-hour cycles to verify proper operation of the timer memory back-up.

4.5.8 Flow meter calibration. With the hypochlorite generation system on and operating normally, measure the discharge flow from one of the dispersal units. Verify accuracy of flow meters by use of a graduated container and stopwatch. Error greater than +/-. 10 percent shall be cause for rejection of the flow meter.

4.5.9 Low flow alarm test. With the hypochlorite generation system on and operating normally, slowly shut seawater inlet valve until the flow meter indicates 20 gpm. At or very near this point, a warning light and audible signal shall come on, and the power supply output shall automatically shut off. Open the inlet valve and return seawater flow to normal. The warning light shall go out and the audible signal shall turn off, and the power supply output shall automatically resume. Failure of either of the tests shall be cause for rejection.

4.5.10 Flow sensors. One at a time, each of the dispersal units with flow sensors shall be immersed in a stream with a known flow rate above 0.5 ft/sec. The flow rate shall be brought down below 0.5 ft/sec at which time the controller should terminate the current to the generator. The flow rate shall be brought back above 0.5 ft/sec at which time the generator should begin to chlorinate again.

4.5.11 Effectiveness of enclosure. The manufacturer shall provide written documentation (see 6.3) proving that the enclosure is rated by "NEMA" to the specified rating.

4.6 Inspection of packaging. Sample packs and the inspection of packaging (preservation, packing, and marking) for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING.

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in Section 2, see 6.9).

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5.1 Packaging requirements. The packaging (preservation, packing, and marking) requirements shall be in accordance with MIL-E-17555 for the level of preservation (A, B, C, or Commercial), the level of packing (A, B, C, or Commercial), marking, and other packaging acquisitioning options therein, as specified (see 6.2).

5.2 Marking. Marking of the system and its components for shipment shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The intended use of the electrochemical hypochlorite generation (EHG) system for dockside service is to control biofouling of ship seawater systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) If first article is specified (see 3.1).
(d) When enclosure is other than specified NAVSEA approval is required (see 3.13).
(e) Level of preservation, level of packing and other packaging options required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.4.1, 4.4.2, 4.4.2(d), (e) 4.4.3(a), (c), (d), (e), (k), (q), (s), appendix A	DI-DRPR-80651	Engineering drawings technical content	----
4.5.11	DI-MISC-80678	Certification/data report	

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Require-

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ments Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Spare parts and tools.

(a) The vendor shall provide a list of recommended spare parts and the specific item to which each part applies.
(b) The vendor should indicate the expected life of the parts requiring replacement and minimum recommended inventory of the spare parts for installation, start-up, continuous operation, and maintenance. Vendor should state whether the recommended spare part is a stock item of a special item, and should furnish name and location of the nearest supplier and appropriate lead time required for delivery.
(c) The vendor should furnish all special tools necessary for installation, start-up, operation, maintenance and adjustment of the equipment and of the accessories furnished by the Vendor.

6.7 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.8 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

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6.9 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.10 Subject term (key word) listing.

Portable
Seawater
Intake flow
Automatic shutdown
Cells

Preparing activity:
Navy - SH
(Project 4610-N130)

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APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. Drawings shall contain the following information:

(a) Master drawings shall be 28 inches by 40 inches.
(b) Manuals shall contain reduced-size copies of master drawings. Therefore scale and lettering size of master drawings shall be large enough so that the drawings are legible when reduced.
(c) Manufacture identification of equipment.
(d) List of descriptive data of the equipment including:

(1) Applicable specification and exceptions.
(2) Complete ratings showing:

(a) Input voltage, frequency, and number of phases.
(b) Output voltage and current at full load.

(3) Duty (if intermittent, indicate duty cycle).
(4) Enclosure classification.
(5) Ventilation.
(6) Type of load.
(7) Ambient temperature range.
(8) Weight of equipment (net).
(9) Notation of the corrosion-resistant treatment of all hardware.
(10) Special features.

(e) List of material, presenting following data in tabular form:

(1) Piece number (for flags of outline drawing).
(2) Name of part (for example, transformer, diode, and other similar parts).
(3) Schematic diagram designation for electrical parts (for example, switch SW1, and other similar designations).
(4) Number of each part required.
(5) Part number or identification assigned by part manufacturer.
(6) National stock number, if known.
(7) Rating of part from part manufacturer's catalog (for example, resistor 10 ohm, 25 watt or similar ratings).

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NOTE: List of material shall exclude ordinary hardware items such as fasteners, component brackets, and other similar hardware items. Heat sink and enclosure assemblies shall be listed.

(f) Outline drawing of equipment showing front, side, top and section views as necessary, to present the following information:

(1) Overall and mounting dimensions.
(2) Location and size of mounting holes.
(3) Approximate arrangement and location of components. Components shall be flagged by piece number.
(4) Location of center of gravity of equipment.
(5) Identification and information plates.
(6) Clearance required for ventilation and to open access doors.
(7) Cable entrance provisions.
(8) Direction of air flow in forced-draft ventilation equipment.

(g) Sketch of terminal boards used for connection of input and output.
(h) Schematic circuit diagram. In preparation of the schematic, emphasis shall be placed in simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram. Each part shall be identified by symbols (R-resistor, L-reactor, and other similar symbols) consistent with connection diagram, list of material and description of operation. Terminals used for connections external to the power supply shall be identified and labeled as to purpose. A note giving input voltage, phases and frequency shall be placed adjacent to input terminals. A note giving output voltage adjustment range and rated output current or power shall be placed adjacent to output terminals.
(i) Connection diagram showing components and wiring in approximately the relative physical positions which they occupy in the equipment. Leads of components, hookup wire lengths used for interconnection, components, terminals and subassemblies shall be marked. Hookup wire size need not be shown. Markings shall be consistent with component descriptions given elsewhere. Notes labeling terminals for external connections shall be provided on connection diagram the same as on schematic diagram.
(j) Description of operation. A brief description of operation of the circuit elements to enable approval authorities and operators to understand circuitry.
(k) Briefly presented operating instructions.
(l) Schematic of piping system.

30.2 Subassembly drawings. Subassembly drawings shall be provided for each subassembly with the following data:

(a) Subassembly name and catalog number or equivalent identification of the item.

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(b) List of material, presenting the following data in tabular form:

(1) Piece number (for flags of outline drawing).
(2) Name of part (for example, transformer, diode, and other similar designations).
(3) Schematic diagram designation for electrical parts (for example, switch SW1, and other similar designations).

NOTE: List of material shall exclude ordinary hardware items such as fasteners, component brackets, and other similar hardware items. Heat sink and enclosure assemblies shall be listed.

30.3 Preliminary drawings. Preliminary drawings representing preliminary preparation of master drawings may be submitted for design approval in any convenient form, provided data are complete.

30.4 Installation drawing. A separate installation drawing showing data of 30.1(a), 30.1(b)(1), 30.1(b)(2), 30.1(d), and 30.1(e) shall be provided. The installation drawing shall include interconnection information of all sub-units of the hypochlorite generating system required for maintenance, subcomponent replacement or repair and to install the unit for service.

30.4.1 Installation drawing. An installation drawing shall be provided showing the interconnections between the hypochlorite generation system and the system to be protected. The drawing shall provide all input power, overboard discharge hypochlorite sensor and seawater connections necessary to place the system in operation. All voltages, flow rates and pressures shall be shown for each connection.

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APPENDIX B

MANUAL TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix details the information that is to be included in a manufacturer's manual for all sub-units of the hypochlorite generation system and includes operating information of the unit as a support system. One manual is to be packaged with each system. This appendix is a mandatory part of the specification.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. MANUALS

30.1 Manuals. When required by the contract or order, manuals shall contain the following information:

(a) Chapter 1 General Information:

(1) Manufacturer's type and catalog or model number.
(2) Navy designation and specification.
(3) National stock number of equipment (if any).
(4) Dimensions.
(5) Weight.
(6) Input power requirements and heat dissipation.
(7) Brief discussion of what the equipment is and what it does.

(b) Chapter 2 Detailed description and principle of operation:

(1) Detailed description of function of each major component. Include theory of operation of components such as power supply, hypochlorite generator, flow sensors, and similar components. Discuss location of each component and provide one or more photographs.
(2) Discussion of how components described in b.(1) function together to provide equipment performance.

(c) Chapter 3 Installation:

(1) Installation instructions (mechanical). Include a sketch. (Reduced size installation drawing may be inserted here.)
(2) Connection instructions (electrical):

(a) Input: give input voltage and current.
(b) Output: give output voltage and current. If receptacles for portable cables are used, note polarity.
(c) Remote control connections, if any.

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(d) Chapter 4 Operating instructions:

(1) Safety precautions.
(2) Start up.
(3) Output current adjustment.
(4) System low flow adjustment.
(5) Ship intake low flow adjustment.

(e) Chapter 5 Maintenance:

(1) Preventive maintenance.

(a) Cleaning.
(b) Inspection.
(c) Lubrication.
(d) Placing in storage and placing in service after storage.

(2) Corrective maintenance.

(a) Troubleshooting data.
(b) Tables and illustrations of measurements of voltage, current, resistance and waveforms at significant test points in the system.

(3) Adjustment after parts replacement

(f) Chapter 6 Parts identification:

(1) Parts tabulation.

(a) Name, piece number and schematic symbol of part.
(b) Equipment manufacturer's identification.
(c) Part manufacturer's identification.
(d) National stock number of part (where applicable).

(g) Chapter 7 Test data:

(1) Design performance data at half and full load, normal input voltage, frequency and 15 ppt salinity water shall be used. Data required are generator current and hypochlorite output in ppm.

30.1.1 Drawings. Reduced-size master and subassembly drawing inserts.

30.1.2 Appendix. Inserts on special components.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the comment number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of this form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-24786(SH)	2. DOCUMENT DATE (YYMMDD) 940321

3. DOCUMENT TITLE
ELECTROCHEMICAL HYPOCHLORITE GENERATION (EHG) SYSTEMS FOR DOCKSIDE SERVICE

4. NATURE OF CHANGE (identity paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC) MR. ANDREW SEELINGER, SEA 03M1 ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: TPOC: 703-602-0213 DSN: 8-332-0213
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND ATTN: SEA 03R42 2531 JEFFERSON DAVIS HIGHWAY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone 703-756-2340 DSN 289-2340

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